Bimlendra Ray

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Education

University of Texas at Dallas — Richardson, TX
Master Of Science in Mechanical Engineering (DSC). GPA: 3.4 — Expected May 2026
Project: Autonomous Vehicles (Current Research Work)
Coursework: Robust Control, Optimal Control, Optimization, Robot Control and Nav.,

Multi-agent system, Modeling and Simulation, MPC, Vehicle Dynamics, Deep learning

Uttarakhand Technical University — Dehradun, India
Bachelor Of Technology in Mechanical Engineering. — 2020
Research work: Review paper on Composite material

NASA Higher Secondary School — Kathmandu, Nepal
Coursework: Physics. — 2016

Projects

End-to-End Autonomous Vehicle Control & Planning System (QUANSER QCAR) — UT DALLAS
Head of QCAR Autonomous Racing Team — Dec 2025 – Present

- Designing and deploying a real-time model-based control architecture (MPC) for autonomous vehicle lateral and longitudinal dynamics on physical hardware
- Optimizing control performance for smoothness, stability, and safety, prioritizing deterministic behavior over end-to-end learning
- PYTHON, OpenCV, TensorFlow, ACADOS

Real-Time Neural Approximation of Constrained MPC via Imitation Learning — UT DALLAS
Feb 2025 – Present

- Designed and implemented a constrained Model Predictive Controller (MPC) for an unstable dynamic system (Linear Inverted Pendulum Model)
- Trained a deep neural network policy using behavior cloning to approximate the MPC feedback law
- Deployed the learned controller in closed-loop simulation and evaluated stability, tracking error, and disturbance rejection
- Analyzed robustness and distribution shift effects in closed-loop learned control
- PYTHON, PyTorch, MPC, Optimal Control, Robotics Simulation

Model Predictive Control for Disturbance-Aware Lap Time Optimization in Autonomous Racing Vehicles using two-step sequential programming — UT DALLAS
Sep 2025 – Dec 2025

- Formulate the lap-time optimization problem under disturbances and model uncertainty.
- Develop an MPC that ensures the actual trajectory stays within a safety region around the nominal optimal racing line.
- MATLAB, SIMULINK, ACADOS

Optimal Trajectory following using AL-ILQR algorithms while avoiding dynamic obstacles using chance constraints UT DALLAS
Feb 2025 – May 2025

- Developed an augmented AL-iLQR algorithm enabling optimal trajectory tracking while safely avoiding dynamic obstacles using chance-constraint formulations
- Modeled a linear quadrotor system, discretized the dynamics using RK4, and applied an LTV approximation to ensure real-time feasibility during constrained trajectory tracking.
- Demonstrated reliable obstacle avoidance performance in simulation; identified and analyzed causes of significant deviation from the reference trajectory to guide future controller refinement
- ILQR, Python

Detecting and Counting the number of objects on Production Line using PLC Wilkes University
Feb 2024 – May 2024

- Developed PLC program and implemented on real hardware (Allen-Bradley)
- Detecting the number of objects in the container and automatically replacing it with empty using PLC controller
- Setting Alarming system to notify the task completion or any fault occurring during process
- PLC

Agri Sort (Sorting Fruits and Vegetable using image classification on moving belt) UT DALLAS
Feb 2025 – May 2025

- Developed a robotic sorting system for fruits and vegetables using ROS and Gazebo.
- Designed and simulated the conveyor belt mechanism.
- Optimized sorting accuracy and reduced manual intervention using AWS Recognition
- Python, ROS, GAZEBO

Experience

Silicon Soft and IT Consultant Pvt. Ltd. Kathmandu, Nepal
Software Developer Jan 2022 – Dec 2023

- Gain confidence in learning and doing the task assigned in time
- Improvised the Journey for Mobile Applications development for the first time in this company. Used Kotlin, Jetpack Compose and .Net framework.
- Worked on a Cross-platform Mobile application with other team members. Used SQL, Kotlin, KMM, KTOR
- Developed Government Web application using REACT and JS
- Handled two project under my responsibilities and led the team member

Gorkha Brewery Pvt. Ltd. (Carlsberg Group) Chitwan, Nepal
Maintenance Engineer (Trainee) Jul 2021– Dec 2021

- Implemented Kaizen to find out the root problem in production line and presented a detailed solution
- Performed extended data acquisition and trend analysis to identify system-level energy losses and recommend efficiency improvements

Leadership & Awards

GORKHA BREWERY PVT. LTD. (PART OF CARLSBERG GROUP) Chitwan, Nepal

Kaizen star of the month November 2021 Jul 2021 –Dec 2021

- Modification of the pull of cap Shute to install empty warning alarm sensor to save beer.
- Used the Kaizen tool to find out the root cause of the frequent stoppage of the filler machine and modified the design. This idea increased production by 0.12 million additional bottles per year.

Skills & Interests

- Programming Language: Python, MATLAB, JavaScript, SQL, PLC
- Simulation: SIMULINK, ANSYS
- Tools/Framework: QUANSER QCAR(AV), ROS, Gazebo, Moveit, Docker, MicroLogix 1100
- Concepts: Controller Design, Multi-agent system, System Modeling and Analysis, Robustness Analysis
- AI/ML: TensorFlow, Deep Learning, Imitation learning
- Design: SolidWorks
- Hardware: Allen-Bradley
- Other Skill: Teamwork, Solo work, Leading project, Presentation